



How Special Robots Can Train Children With Autism

Autism is increasing at a tremendous rate among children in India. Children with autism suffer from varied deficiencies in skills and learning disabilities. It's important to detect it at early stages, or else, children might suffer for the rest of their lives

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One of the most worrying aspects among children in India is the increase in the incidence of autism in recent times. From what was thought to be one in 500 children having autism, today, the number is standing at a staggering one in 68 children. To compound the issue, 40 per cent of children with autism don't speak, and this leads to chances of non-detection or late detection. Autism is visible from the age of one to three years of a child's life. In a few instances, it is visible even earlier. If autism is not detected at early stages and addressed, children suffer later in their lives. Some of the myths and the reality related to autism is shown in the box below.

Defining autism

Treating autism is effective when it is detected earlier in a child. Some of the symptoms that a child with autism exhibits are:

- Sticks to a set of behaviours and resists change in daily activities
- Struggles to make sense out of the day to day activities
- Gathering of people at events and places confuses the child and causes anxiety

In general, these are called 'Triads of Impairments.' The level at which autism spreads across a spectrum varies from person to person. Evaluating the children with a country-specific evaluation process will help in identifying the level to a large extent. One key aspect that we need to keep in mind is that 80-90 per cent brain growth happens in childhood, so unless the detection and training are done in the early stages, things can become more difficult.

Impact of autism

Children with autism suffer from varied deficiencies in skills and learning disabilities. In simple language, autism interferes with the child's:

- Social interactions
- Social communication
- Activities and interests (social imagination)

These impairments are typically explained by a simple graphic shown in Fig. 1.

One of the most important skill sets that mostly affect the children is psychomotor skills. Psychomotor learning is the relationship between cognitive functions (consciously involved in functioning) and physical movement. Psychomotor learning

*Myth 1 – Autism is not prevalent in India
It is highly prevalent and growing*

*Myth 2 – No one knows what causes it
The problem lies in the way brain cells connect*

*Myth 3 - Therapy can't 'rewire' the autistic brain
Therapy can open new neural pathways of learning*

*Myth 4 - Autism is not genetic
There is clearly a genetic angle to autism*

*Myth 5 - Autism cannot be reversed
It has been reversed on animals in labs*